Exploring Virtualization Efficiency, Security Enhancements, and Scripting for File and Operating Systems*

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1 Virtualisation Efficiency Benefits

1.1 Understanding virtualization

Virtualization in simple terms is the sharing of physical resources between multiple virtual machines. You can think of it as a "computer within a computer, implemented in software." [5] This is done by creating an abstraction layer over the physical hardware of the host machine so multiple computers (Virtual Machines) can run on one hardware. They can also share resources such as network cards, CPU, memory, Physical storage and others. Virtual machine is a virtual depiction of machine hardware which is maintained by a virtual machine monitor (VMware Fusion, Oracle VM VirtualBox). (Abstraction diagram here[create/find on internet])The concept of virtualization isn't new it has been here since 1960. But with the introduction of X86 Architecture and platforms, it now has been in the sighs of many companies because it provided simple seamless implementation without significant interruption to operations and low acquisition. [6] [5]

1.2 Benefits of Virtualization

The benefits of virtualization are considerable. Virtualization can help with security, scalability, cost cutting, efficient uses of resources and other. Non virtualized systems can be inefficient when non virtulzed resources aren't in use and the hardware is sitting idle without any computing task ready to run. While in virtulized system once resources become available hypervisor can allocate them accordingly to another task so the physical resources are always in use. With this comes another benefit of security and corruption prevention where in a normal non virtulized system the security and backup lay on one specific machine and user. The virtualized system security depends on the security of the whole system. This can lead to better and more secure disaster prevention where we can back up each VM, so in case of a natural disaster, there isn't any loss of data. In simple terms, virtualization helps with the growth scaleability of an organization.

For ACME Games virtualization will not only help with cost cutting and security but also with:

1. Minimize Servers
2. Reduce Hardware
3. Quick Redeployment
4. Simpler Backups
5. Better Testing

[5] [4]

1.3 Cost of virtualization and energy efficiency

This part delves into total cost of ownership for virtulized systems compared to standard systems and energy efficiency of medium to large organizations. We will look in to direct and indirect costs of virtualization for organizations. With real life examples of companies that went through restructuring of their servers to implement virtualization.

1.3.1 Cost of virtualization

First we need to define what are the major direct cost categories for organization to implement virtulized system.

1. Hardware cost

   Represented as expenses that are expected to be bought so virtualization can be implemented to the system. These include physical technologies that would allow to create new virtulized system [5]

2. Software cost

   Are expenses derived from buying and/or licensing software which allows access to virtual machines and all its possible functions. [5]

3. Deployment costs

   These costs are other expenses not directly associated with the concept of virtualization. Costs like disruption to existing operations and space improvement.

These are direct costs of implementing virtualization they can be one-time or recurring. These expenses represent the general figure for medium to large firms. These can be cut down further with the right licensing agreement and the right negotiation with hardware providers.

There are also second indirect costs for virtualization

1. Training personnel cost
2. Maintenance cost
3. Energy cost

Hardware cost The initial virtualization cost can be considered high when a classical system is already in place and working. Let's consider the Hardware cost of implementing new servers. We will look into the 2 most popular servers Which are Dell's PowerEdge R6625 and PowerEdge R7625. With these 2 servers, Dell allows extensive customization to get the best price for the best performance and energy efficiency for specific tasks. The table below shows the specifications of each server and the lowest configuration cost per 1 server.

Dell PowerEdge		
Name:	R7625	R6625
Processor	2X AMD EPYC 9224	
Memory (DDR5 DIMM slots/max)	24(1.5TB)	
Disk Drivers up to:	8 X 3.5	4 x 3.5
	12 X 3.5	8 x 2.5
	8 X2.5	10 x 2.5
	16 x 2.5	2 x 2.5
	24 X 2.5	
NVMe driver up to	24	10
Gen 5 PCIe slots up to:	4	2
Gen 4 PCIe slots up to:	8	3
Accelerator support	2 x 300 W DW or 6 x 75W SW	3 x 75 W SW
Lowest configuration Cost	28124.8 USD	8399 USD

Figure 1: Table comparesen made by author. Sourced: [1] [3] [2]

Software cost There are multiple virtualization software to consider. We will look into 2 different virtualization software VMware vSphere and Citrix XenServer.

1. VMware vSphere

   Offers 2 different options vSphere standard and Foundation.

   vSphere Foundation is their flagship virtualization software which allows access to the full set of virtualization features. This software is for medium to large companies. The set price as of February 2024 is 135 USD per CPU Core.

   For one of the above-mentioned server, it would come to 7200 USD Yearly

   vSphere Standard is aimed at small to medium-sized companies. It offers essential virtualization capabilities. vSpahere standard is priced at 50 USD per CPU core.

   For one of the above-mentioned server, it comes to 2400 USD Yearly

2. Citrix XenServer

   Also Offers Standard and Premium Edition. Unlike vSphere, XenServer Operates on a cost per-CPU socket.

   Their standard edition is entry-level virtualization software for lower cost and organizations can still benefit from Technical Support and Maintenance. Nevertheless, the Premium version offers GPU virtualization, Dynamic Workload Balancing, Thin provisioning for shared block storage devices and other features which are essential to ACME Games.

   The premium version for 1-4 Sockets is 875 USD per socket per year. The standard version price for 1-4 sockets is 440 USD per socket per year.

   The price goes down when more sockets are licensed.

1.3.2 Energy efficiency

The cost of Energy is hard to predict due to constant economic changes. The cost of energy changes rapidly each month. However from research done by Jukka Kommeri, Tapio Niemi and Olli Helin in their Energy Efficiency of Server Virtualization from 2012 [6] shows that with correct implementation virtualization can save energy. In their words "research indicates that idle power consumption of virtualized server is close to zero" [6].

Real world example As research into virtualization progresses even more we can expect that energy consumption will lower. Case study done in 2010 by K. Thirupathi Rao,P. Sai Kiran and L.S.S.Reddy(Energy Efficiency in Datacenters through Virtualization: A Case Study [7]) indicetes that after virtualization datacenter obtained power improvement of 65 percent furthermore data center reduced their number of servers from 500 to 140. Practical investigation in to Croatian company done in 2012 that went through virtualization is the best example for ACME Games. They used Dell PowerEdge 2970 servers for their virtualization. Figure 2 shows lower numbers of servers with predicted growth of 0.25 server per year. [5]

Description	Value	Source
Present number of physical servers	7	Interview
Number of physical servers – post virtualization	2	Own source
Life time of server	4 years	Industrial best practice
Chosen model of server	Dell PowerEdge 2970	Own source
Number of virtual machines per physical server	4	Industrial best practice
Annual growth of servers	0.25 servers per year	Interview

Figure 2: Table from: [5]

Their virtualized system yields economic energy efficiency of 72.41 percent. The energy cost of their non-virtualized system per year was 45,439.69 USD(adjusted to inflation as of 2024) while their virtualized system 12,535.19 USD (Adjusted to inflation as of 2024). This is 27.59 percent less energy consumed per year.

Overall the economic effect of virtualization for the company was 57.63 percent.

There it is safe to assume that with a correct implementation of virtualization, ACME Games can save on energy consumption, resources and capital.

2 Security Issues

Any technological enterprise's biggest concern is the mitigation of unauthorised access to company data. This unauthorised access can be within the company's corporate hierarchies or a malicious attack by a third party. While access permission within a company's structure is achieved by setting up correct entry dispensation and secured data movement infrastructure within the company's internal systems, third-party attacks are harder to prognosticate, therefore establishing robust threat assessment and a durable security panel is exceedingly important compared to other parts of virtualization.

2.1 Operating system security issues

In this day and age, we can find operating systems everywhere. This leads to day-to-day exponential improvement in developing malicious algorithms to infect our Operating system. Multiple common vulnerabilities can be exploited

1. Buffer Overflow
2. Code Injections
3. Privilege Escalation
4. Denial of Service
5. Zero-Day

A great example is Microsoft's vulnerability issue from 2023 which was patched in February 2024 and tracked under code CVE-2024-21338. This issue affected the kernel mode in different versions of Windows 10 and 11. This exploit used CWE-822 Weakness where the attacker supsedies internal pointer for memory location which wasn't expected. If the pointer dereferences in to write the attacker can start modification of variables or carry out code execution.On the other hand, dereferencing pointer to read allows the reading of sensitive data. This issue falls into the Zero-Day exploit. This CWE-822 issue can be exploited in any language that uses pointers(c/c++, assembly...)

This part explores the most common operating system security issues.

Buffer overflow occurs when more information is written to the buffer beyond its memory. Buffer overflow attacks on a system aim to modify the state of memory so that a third party can execute code that allows it to take over the machine. This is done by precisely inputting data so that the buffer would copy this information to the adjacent location in memory.

Code Injections is security exploitation where a program misinterprets input data as part of its code. Figuratively the attacker injects code into the run of the program and changes its behavior to gain access to the machine.

Privilege Escalation is accomplished by exploiting systems or programs' vulnerability to gradually gain access to different resources and modes of the operating system. This way attackers can go from user mode with the least amount of privilege to the kernel mode to change the behavior of the OS.

Denial of Service attacks are meant to disrupt the network for the attended user so they cannot use it. In these attacks third-party floods the network with a large amount of requests which it can't handle essentially shutting the network down.

Zero-Day exploitation of a software issue that developers don't know about. This isn't any specific common issue being exploited. In this scenario, attackers are trying to find software vulnerabilities faster than developers can find them.

2.2 Virtualization security issues

The benefits of virtualization on security are vast. As virtualization progresses, new research into virtualized systems issues emerges. Although this research suggests a new way of thinking about security, attackers also expand their roster of knowledge. That's why it is impossible to cover every security issue.

There are two ways of thinking when creating threat assessment in virtualization one would be to consider hypothetical in many cases like hardware virtualization where many of the physical components are made in countries with no regard to security. In this case, we haven't seen any considerable foreign intervention in the production of hardware. Therefore we are going to look into the real threats in virtualization.

2.3 Virtualization threat assessment

For a virtualized system we identify three major routes that third parties can exploit.

1. Hypervisor-based Attacks
2. VM-based Attacks
3. VM Image Attacks

As shown in the figure, all these branch out into another subset of specific malicious intrusions. We are going to look into the most common Virtualization attacks over the years and explain briefly how we can prevent such issues from arising.

2.3.1 Most common security risks in virtualized environment

VM Sprawl Virtual machine sprawl occurs when an unmanaged number of VMs are created for one specific task and forgotten about. This leads to a waste of resources, whereby already created VMs take up resources that aren't in use and can be utilised somewhere else. Also, prolonged unchecked VMs can lead to loss of sensitive data, or by non-updating and managing VMs, vulnerabilities can be created to access the inner system. This can be mitigated by creating VM lifecycle insurance or by strict laws under which can VMs be created, managed and terminated

Malware attacks VMs are as any other computing technology susceptible to malware attacks. This can be a bigger danger in a virtualized environment when users of VMs don't have proper security training which can lead to the infection of one VM. This wouldn't be as big of an issue as it is in a normal system but in virtualization, this can lead to the spread of malicious software to the host machine and all other VMs that run on the same physical machine. This can be mitigated by setting up Firewalls, Host intrusion prevention systems and anti-virus/spyware

3 File System Scripting

4 Operating System Scripting

5 Conclusions

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